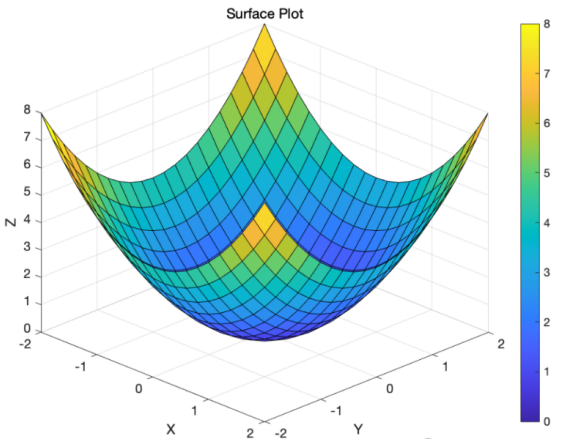
美赛O奖干货|Matlab绘图归纳整理

> 2025年美赛倒计时四周！今天给大家分享**如何用matlab绘制高大上图片的技巧**，帮助大家在论文中的图标展示更有创新和创意点，吸引评委老师的眼球！！！

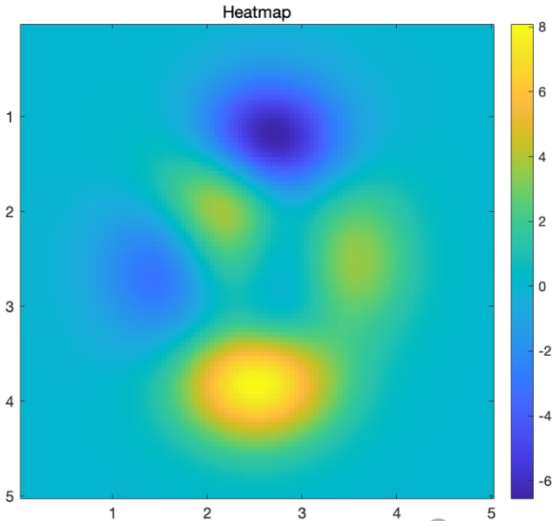
1.surf函数：绘制三维曲面图

```
%创建数据
x= -2:0.2:2;
y= -2:0.2:2;
[x,Y]= meshgrid(x,y);
Z = X.^2 + Y.^2;
%绘制三维曲面图
surf(x,Y, Z);
% 添加标题和杭签
title('surface Plot');
xlabel('x');y
label('y');
zlabel('z');
% 添加颜色栏
colorbar;
% 设置视角
view(45, 30);
```



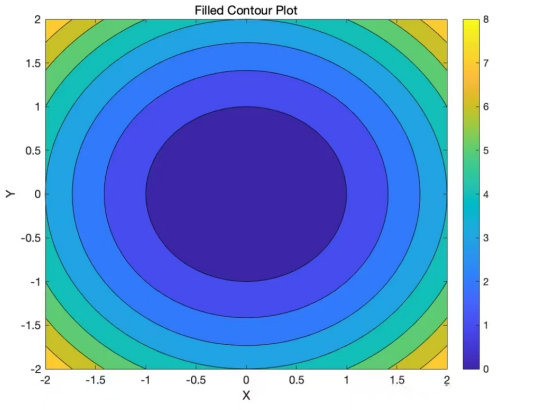
2.imagesc函数：绘制热图

```
% 创建数据
data = peaks(100);
% 绘制热图
imagesc(data);
% 添加标题和颜色栏
title('Heatmap');
colorbar;
% 调整坐标轴
axis image;
% 添加刻度标签
xticks(8:20:100);
yticks(0:20:100);
% 添加刻度标签标签
xticklabels({'0', '1','2','3'
yticklabels({'0', '1','2','3'
```



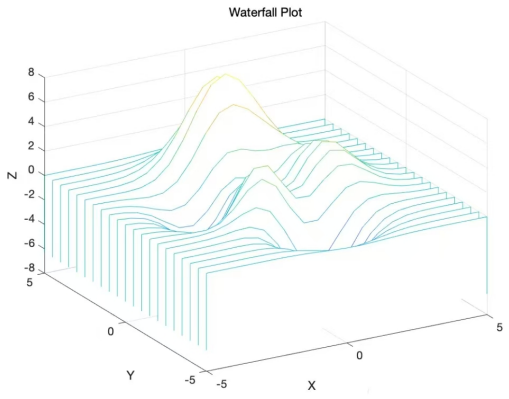
3.contourf函数：绘制填充等高线图

```
创建数据
x= -2:0.1:2;
y= -2:0.1:2;
[x,Y]= meshgrid(x,y);
Z = X.^2 + Y.^2;
%绘制填充等高线图
contourf(X,Y, Z);
% 添加标题和标签
title('Filled contour plot')
xlabel('x');
ylabel('y');
% 添加颜色栏
colorbar;
```



4.waterfall函数：绘制瀑布图

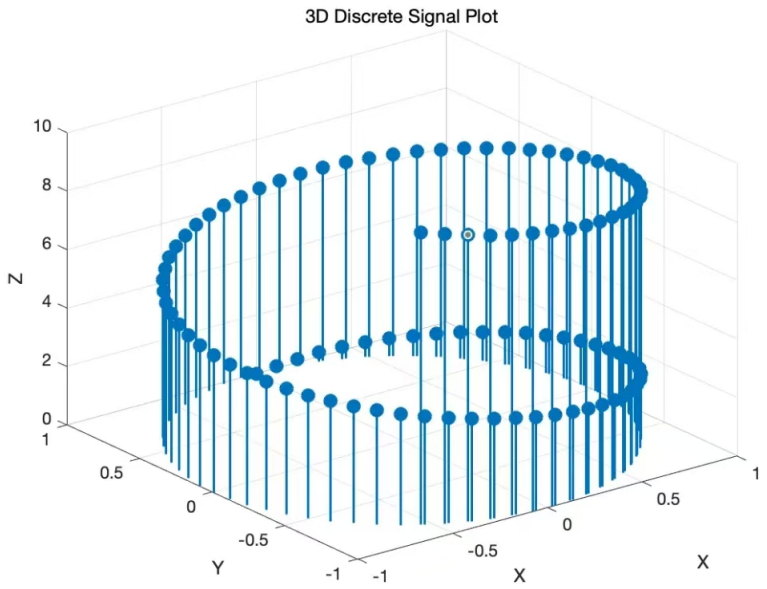
```
%创建数据
x= -5:0.5:5;
y= -5:0.5:5;
[x,Y]= meshgrid(x,y);
Z = peaks(length(x));
%绘制瀑布图
waterfall(X,Y，Z);
% 添加标题和栿签
title('Waterfall Plot');
xlabel('x');
ylabel('y');
zlabel('z');
% 设置视角
view(-30，30);
```



5.stem3函数：绘制三维离散信号图

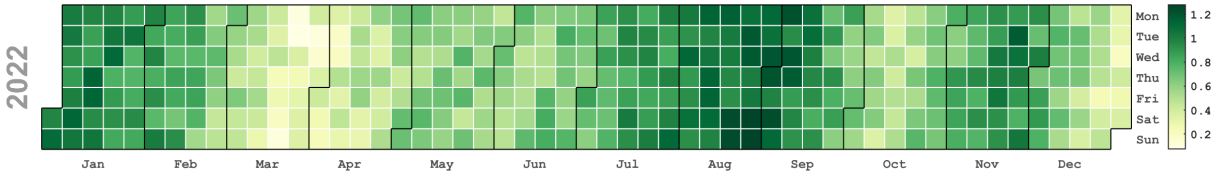
```
% 创建数据
t= 0:0.1:10;
x= sin(t);
y= cos(t);
z = t;
% 绘制三维离散信号图
stem3(x,y,z,'filled','linewidth',1.5,'Markersize',8);

% 添加标题和标签
title('3D Discrete signal Plot');
xlabel('x');
ylabel('y');
zlabel('z');
% 添加网格
grid on;
```



6.绘制日历热力图并调整大小

```
% 绘制日历热图
heatmapDT(2022,T,V)
% 调整图窗和坐标区域大小
set(gcf,'Position',[100,500,1500,260])set(gca,'Position',[.03
```



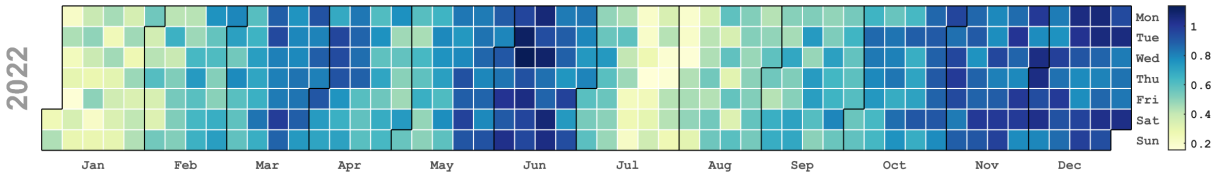
修改colormap

```
colormap(pink)
```

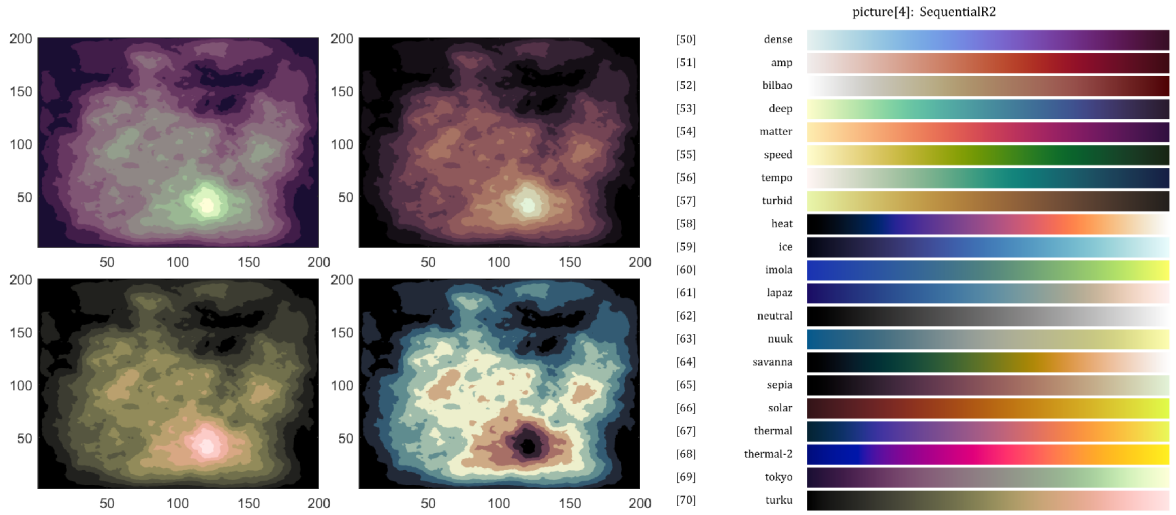
自己定义点颜色：

```
CM=[1.0000     1.0000    0.8510
    0.9487     0.9800    0.7369
    0.8617     0.9458    0.6995
    0.7291     0.8938    0.7109
    0.5237     0.8139    0.7308
```

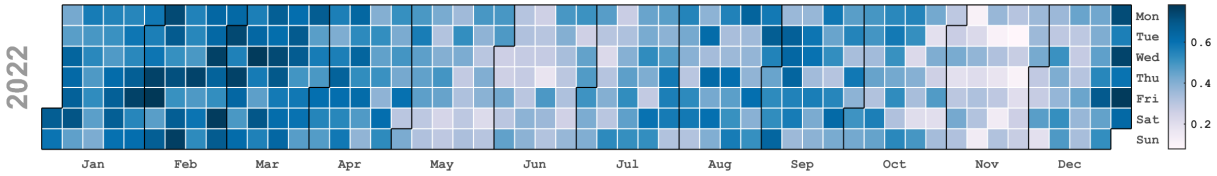
```
    0.3433    0.7465    0.7558
    0.2036    0.6610    0.7629
    0.1155    0.5504    0.7444
    0.1298    0.4050    0.6759
    0.1398    0.2788    0.6160
    0.1141    0.1793    0.5162
    0.0314    0.1137    0.3451】;
CMX=linspace(0,1,size(CM,1));
CMXX=linspace(0,1,256);
CM=[interp1(CMX,CM(:,1),CMXX,'pchip'), ...
interp1(CMX,CM(:,2),CMxx,'pchip'),...
interp1(cMx,CM(:3),cMxx,'pchip')];
colormap(CM)
```

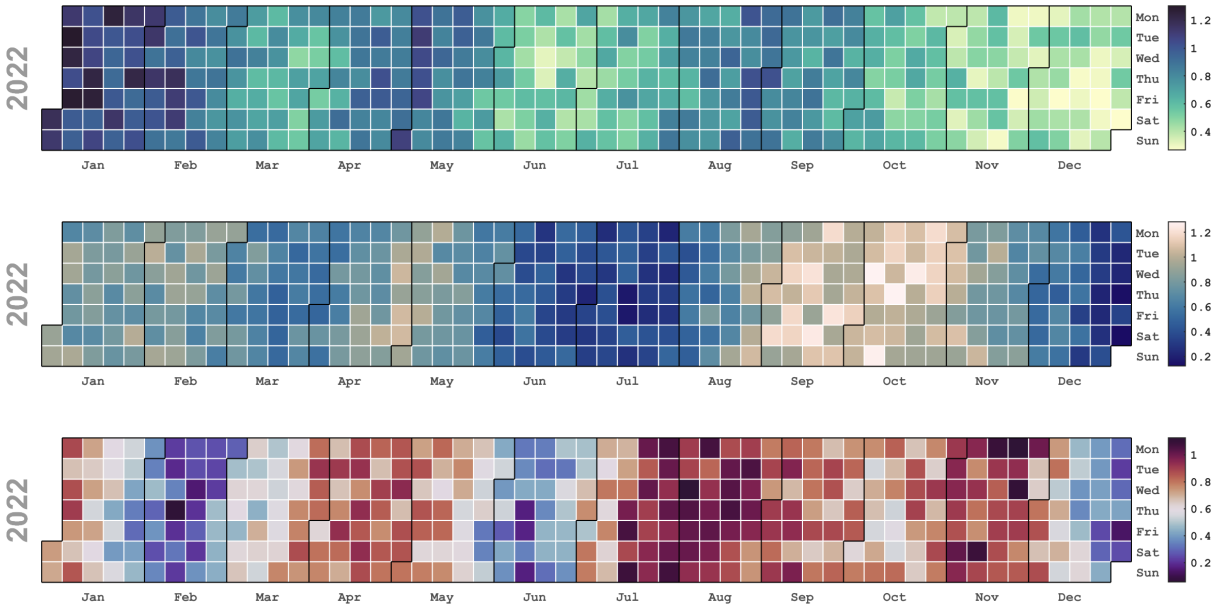


可以配合slanCM配色工具



使用效果



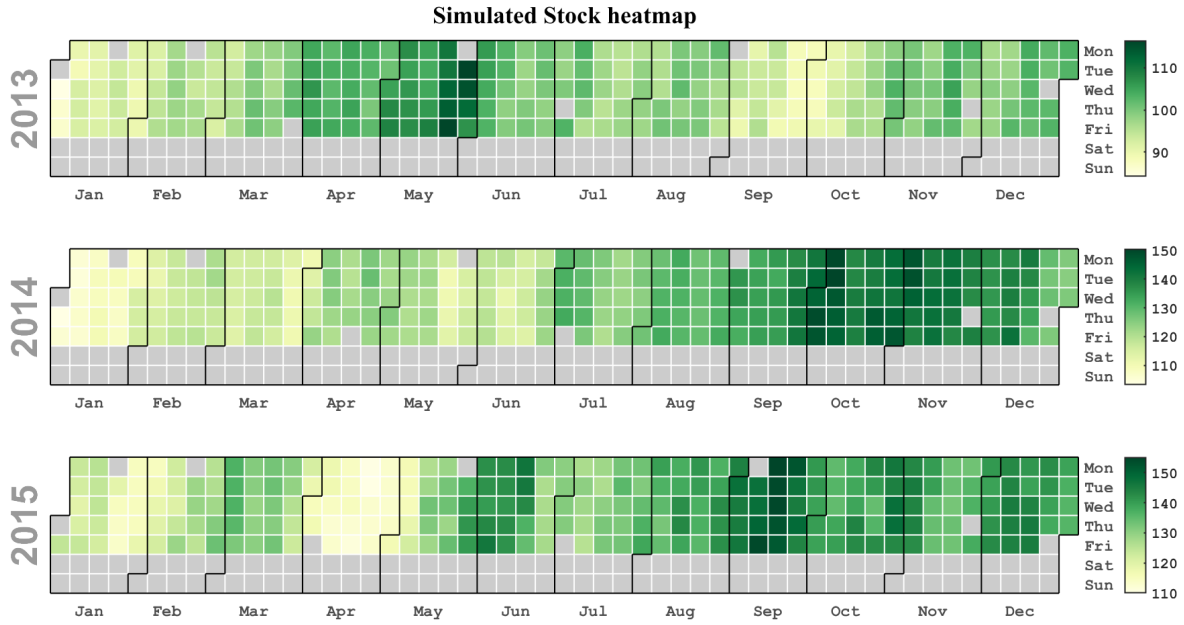


7.多子图

> 就将第一个参数设置为坐标区域即可,以下不用随机生成的数据了，用一个matlab Financial Toolbox工具箱自带的数据：

```
load5imulatedstock.mat
T=TMW.Time;
V=TMW .High;
% 绘制三个热力图
ax1=axes(gcf,'Position',[.06,2/3+.01,1-.07,1/3-.03]);heatmapD

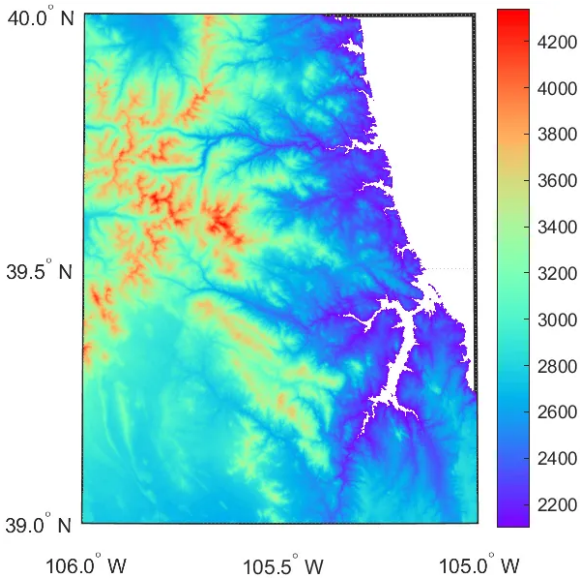
ax2=axes(gcf,'Position',[.06,1/3+.01,1-.07,1/3-.03]);heatmapD
ax3=axes(gcf,'Position',[.06,0+.01,1-.07.1/3-.031);heatmapDT(
% 绘制标题并调整位置
TT=title(ax1,{'simulated Stock heatmap'},'fontsize',18, FontW
TT.Position(2)=-.1;
set(gcf,'Position',[100,100,1200,620])
```



8.带NaN图例的地图绘制

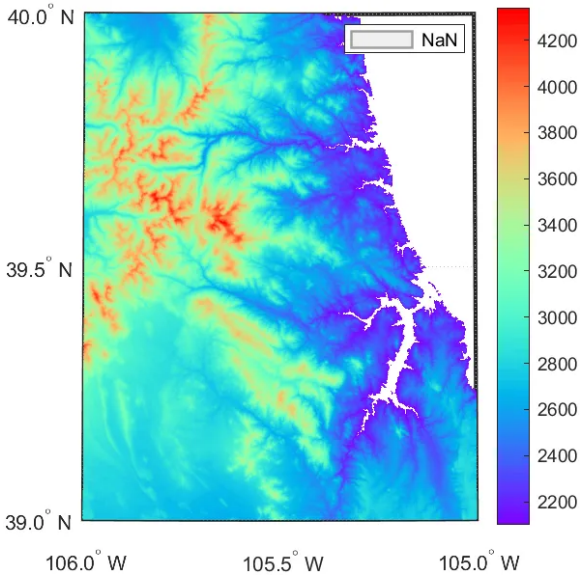
> ### 添加NaN
>
> 对于这个工具箱的这种绘图方式，比如我们想将数值低于2100的数值改成NaN,首先将原始代码改成这样:

```
%需要napping toolbox
[Z,R]=readgeoraster('n39 w106 3arc_v2.dt1','0utputType','doub
key.GTModelTypeGeoKey=2;
key.GTRasterTypeGeokey=2;
key.GeographicTypeGeoKey=4326;
filename='southboulder.tif';
% 设置NaN值
Z(Z<2100)=nan;
geotiffwrite(filename,Z,R,'GeoKeyDirectoryTag',key)
usamap([39 40][-106 -1051)
ggeoshow(filename,'DisplayType','mesh');
%190 300 363
colormap(nclCM(15,100))
cbHdl=colorbar();
```



生成NaN图例

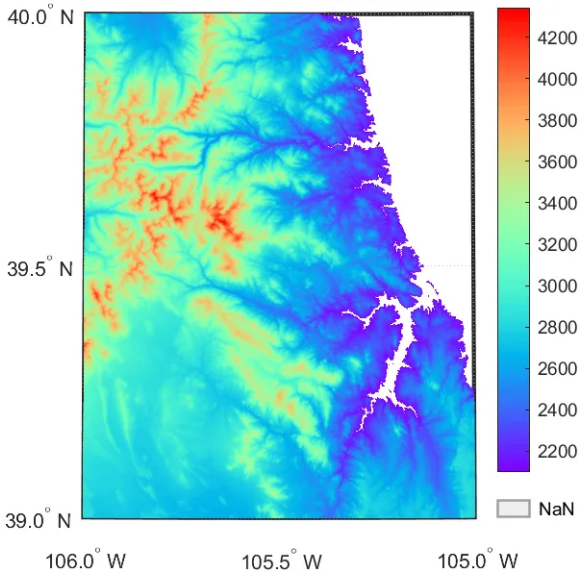
```
% 绘制han图例
nanHdl=fi1l([0,1,1,0]-1000,[0,0,1,1]-1000,[240,240,240]./255,
     'EdgeColor','none','Edgecolor',[160,160,160]./255,'LineW
     'DisplayName',’ NaN');
lgdHdl=legend(nanHdl);
```



修改图例位置

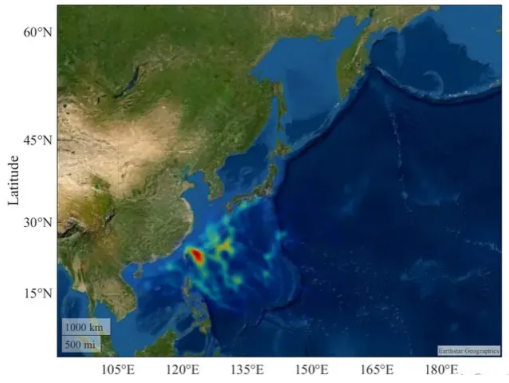
```
% 修改colorbar位置
tPosition=cbHdl.Position;
```

```
cbHdl.Position(1)=cbHdl.Position(1)+tPosition(3).*1.5;
cbHdl.Position(2)=cbHdl.Position(2)+tPosition(4)./10;
cbHdl.Position(4)=cbHdl.Position(4)-tPosition(4)./10;
% 绘制nan图例
nanHdl=fi11([0,1,1,0]-1000,[0,0,1,1]-1000,[240,240,240]./255,
     'EdgeColor','none','EdgeColor',[160,160,160]./255,'lineW
     'DisplayName',' NaN');
lgdHdl=legend(nanHdl);
lgdHdl.Box='off';
lgdHdl.ItemTokenSize=[16,16];
lgdHdl.Position(1)=tPosition(1)+tPosition(3).*1.3;
lgdHdl.Position(2)=tPosition(2);
```



9.地图绘制

```
load cycloneTracks
geodensityplot(...
cycloneTracks.Latitude,...
cycloneTracks.Longitude,...
cycloneTracks.Windspeed,...
'FaceColor','interp')
```



📌 作图规范：

- 线宽合适，默认的太细，最好是1.1；
- 字体调大，默认的太小，最好是16；
- 散点内部填充颜色要好看，不要刺眼，要手动调RGB值设置颜色；
- 坐标轴边框也太细，一定要加粗到1.1；
- 作图一定要加上横纵坐标描述；
- 显示出坐标轴范围要留出空隙，不要让图像紧靠边框，且上下左右空隙相等。